

Convergence of Chebyshev's Method for $z(z^{18} - 1)$

Xiaojian Zeng

PBC School of Finance, Tsinghua University, 100190

xiaojianzeng@foxmail.com

Abstract

For the polynomial $p_{18}(z) = z(z^{18} - 1)$, we prove that every Fatou component of its Chebyshev map is contained in an attracting basin of one of the nineteen roots. The map is therefore convergent in the root-finding sense. The map-specific step is an exact certificate: after two iterates, every free critical point enters a real invariant interval on which the induced map contracts strictly toward the origin. A complete critical-orbit classification, the classification of periodic Fatou components, and Sullivan's no-wandering-domain theorem then give the global conclusion. The result concerns only the first even parameter not covered by the previously supplied general theorem, namely $n = 18$; it makes no claim for another even parameter or about Julia-set topology or convergence speed.

1 Statement and context

Let

$$F(z) = C_{18}(z) = \frac{z^{19}(12654z^{36} - 684z^{18} - 306)}{2(19z^{18} - 1)^3}. \quad (1)$$

The map extends rationally to the Riemann sphere $\widehat{\mathbb{C}}$. Write $\mathcal{F}(F)$ and $\mathcal{J}(F)$ for its Fatou and Julia sets. For a fixed point r , let

$$B(r) = \{z \in \widehat{\mathbb{C}} : F^m(z) \longrightarrow r\}.$$

Following the root-finding convention of McMullen [2] and Nayak–Phogat [3], the map F is *convergent* if every Fatou component is contained in one of the basins corresponding to a root of the input polynomial.

Nayak and Phogat study the family associated with $p_n(z) = z(z^n - 1)$. Their Theorem D proves convergence for $n \leq 16$ or odd n , and their concluding question asks about even $n \geq 18$. For $n = 18$, their table records a floating-point critical orbit that appears to enter the basin of the origin [3]. The purpose of this note is to replace that numerical observation at this one parameter by exact capture and to prove the required global Fatou-set statement.

Theorem 1. *Every connected component of $\mathcal{F}(F)$ is contained in $B(r)$ for some*

$$r \in \{0\} \cup \{z \in \mathbb{C} : z^{18} = 1\}.$$

Equivalently, the map C_{18} in (1) is convergent in the root-finding sense.

The proof has two logically separate parts. Sections 2 and 3 give a rational-arithmetic certificate for every free critical orbit. Section 4 checks all remaining critical points and exhausts the possible periodic Fatou components. The deliberate use of strict rational inequalities is methodologically related to the proof-carrying inequality certificates in [5] and, in a complex-analytic setting, [4]. Those papers supply no result about rational dynamics; the dynamical inputs used here are the standard results recalled in [1, 3].

2 An exact free-critical orbit

Put

$$Q(x) = \frac{x^{19}(12654x^{36} + 684x^{18} - 306)}{2(19x^{18} + 1)^3} = xq(x^{18}), \quad q(t) = \frac{t(12654t^2 + 684t - 306)}{2(19t + 1)^3}. \quad (2)$$

Let $\lambda = e^{\pi i/18}$. Direct substitution in (1) gives

$$F(\lambda^k x) = \lambda^k Q(x) \quad (x \in \mathbb{R}, k \text{ odd}). \quad (3)$$

Differentiating (1) and cancelling common constants gives

$$F'(z) = \frac{18 \cdot 19 z^{18}(z^{18} - 1)^2(703z^{18} + 17)}{2(19z^{18} - 1)^4}. \quad (4)$$

Thus the eighteen free critical points are

$$\lambda^k \tilde{c}, \quad k \in \{1, 3, \dots, 35\}, \quad \tilde{c}^{18} = t_0 := \frac{17}{703}.$$

Equation (3) reduces all of their orbits to the real Q -orbit of $\tilde{c}$.

Let $x_0 = \tilde{c}$, $x_{j+1} = Q(x_j)$, and $t_j = x_j^{18}$. Then

$$t_{j+1} = t_j q(t_j)^{18}. \quad (5)$$

Lemma 2 (Second-iterate certificate). *The first two steps in (5) satisfy*

$$\frac{16}{125} < t_1 < \frac{13}{100}, \quad -\frac{1}{50} < q(t_1) < 0, \quad 0 < t_2 < \frac{1}{100}.$$

Proof. Exact reduction gives

$$q(t_0) = -\frac{10693}{9747}, \quad t_1 = \frac{17 \cdot 10693^{18}}{703 \cdot 9747^{18}}.$$

The two relevant integer cross-products are

$$\begin{aligned} D_- &= 125 \cdot 17 \cdot 10693^{18} - 16 \cdot 703 \cdot 9747^{18} \\ &= (6\,509\,250\,070\,784\,751\,692\,384\,879\,424\,159\,087\,472)10^{36} \\ &\quad + 754\,312\,233\,670\,749\,508\,990\,922\,543\,077\,280\,853 > 0, \\ D_+ &= 13 \cdot 703 \cdot 9747^{18} - 100 \cdot 17 \cdot 10693^{18} \\ &= (83\,439\,327\,875\,066\,059\,753\,403\,827\,793\,076\,796\,853)10^{36} \\ &\quad + 295\,339\,472\,208\,955\,993\,979\,384\,424\,711\,909\,671 > 0. \end{aligned}$$

They prove the asserted lower and upper bounds for t_1 .

Let $A(t) = 12654t^2 + 684t - 306$. This polynomial is increasing for $t > 0$, and

$$A(13/100) = -\frac{16137}{5000} < 0.$$

Since the remaining factors in $q(t)$ are positive for $t > 0$, it follows that $q(t_1) < 0$. For the other inequality, expansion gives

$$q(t) + \frac{1}{50} = \frac{H(t)}{50(19t + 1)^3}, \quad H(t) = 323209t^3 + 18183t^2 - 7593t + 1. \quad (6)$$

At the left endpoint,

$$H(16/125) = \frac{9423189}{1953125} > 0, \quad H'(16/125) = \frac{202315887}{15625} > 0.$$

Moreover, $H''(t) = 1939254t + 36366 > 0$ for $t \geq 0$. Hence H is positive throughout $[16/125, 13/100]$, and (6) gives $q(t_1) > -1/50$. Finally, by (5),

$$0 < t_2 = t_1 q(t_1)^{18} < \frac{13}{100} \left(\frac{1}{50} \right)^{18} < \frac{1}{100}.$$

□

3 The exact capture threshold

Define

$$P(t) = 13186t^3 + 1425t^2 - 96t + 1.$$

Lemma 3 (Invariant interval). *The polynomial P has a smallest positive root t_* , and*

$$\frac{1}{100} < t_* < \frac{1}{70}.$$

If $0 < |x|^{18} < t_*$, then $Q^m(x) \rightarrow 0$.

Proof. The derivative factors as

$$P'(t) = 6(6593t^2 + 475t - 16).$$

The quadratic in parentheses is increasing on $[0, \infty)$, and

$$6593(1/100)^2 + 475(1/100) - 16 = -\frac{105907}{10000} < 0.$$

Therefore P decreases on $[0, 1/100]$. The exact signs

$$P(1/100) = \frac{97843}{500000} > 0, \quad P(1/70) = -\frac{1808}{42875} < 0$$

show both that P has no zero up to $1/100$ and that it has a positive zero before $1/70$. This proves the root isolation.

Two expansions give

$$q(t) + 1 = \frac{P(t)}{(19t + 1)^3}, \quad q(t) - 1 = -\frac{532t^3 + 741t^2 + 210t + 1}{(19t + 1)^3}. \quad (7)$$

Since $P(0) = 1$ and t_* is its first positive zero, (7) implies

$$-1 < q(t) < 1 \quad (0 < t < t_*).$$

Consequently,

$$|Q(x)| = |x| |q(|x|^{18})| < |x| \quad (0 < |x|^{18} < t_*), \quad (8)$$

and the interval in (8) is forward invariant. Thus $|Q^m(x)|$ decreases to a limit $L \geq 0$. If $L > 0$, continuity would give

$$L = L|q(L^{18})| < L,$$

a contradiction. Hence $L = 0$.

□

Lemmas 2 and 3 give $t_2 < 1/100 < t_*$, so $Q^m(\tilde{c}) \rightarrow 0$. Equation (3) now proves the following map-specific conclusion.

Proposition 4. *Every free critical point of F belongs to the full attracting basin $B(0)$.*

The proposition concerns the full basin, not necessarily the immediate component containing the origin. This distinction is important because the target statement quantifies over every Fatou component.

4 Exhaustion of the Fatou set

The numerator and denominator in (1) are coprime and have degrees 55 and 54, respectively, so $\deg F = 55$. Equation (4), together with the local degree at a pole, gives the complete critical list:

1. 0, with critical multiplicity 18, is a superattracting fixed root;
2. the eighteen roots of unity, each with critical multiplicity 2, are superattracting fixed roots;
3. the eighteen free critical points $703z^{18} + 17 = 0$, each simple, belong to $B(0)$ by Proposition 4;
4. the eighteen poles $19z^{18} = 1$, each of local degree 3, map to ∞ .

The corresponding critical multiplicities sum to

$$18 + 18 \cdot 2 + 18 + 18 \cdot 2 = 108 = 2 \deg(F) - 2,$$

so neither a finite point nor infinity has been omitted. Directly from the leading terms,

$$F(z) = \frac{333}{361}z + O(z^{-17}) \quad (z \rightarrow \infty).$$

The multiplier at infinity, in the coordinate $w = 1/z$, is therefore $361/333 > 1$. Hence infinity is repelling, and its pole preimages lie in the Julia set.

We use three standard facts for rational maps of degree at least two [1, 3]: every Fatou component is periodic or preperiodic; a periodic component is an attracting domain, a parabolic domain, a Siegel disk, or a Herman ring; and an immediate attracting or parabolic basin cycle contains a critical point, while the boundary of a rotation domain lies in the closure of the postcritical set.

There is no non-root attracting or parabolic basin. Indeed, a critical point in its immediate basin cycle would have to converge to that cycle. The complete list above makes this impossible: the root critical points are fixed roots, the free critical points tend to 0, and the pole critical points land at the repelling fixed point ∞ .

There is no rotation domain. Let

$$\mathcal{P}(F) = \{F^m(c) : c \text{ is a critical point of } F, m \geq 0\}.$$

Its closure is a finite union of convergent free-critical orbits, fixed root orbits, and finite pole-infinity orbits. All free-critical orbit points and all roots lie in the Fatou set, and the only limit of each free-critical orbit is 0. It follows that

$$\overline{\mathcal{P}(F)} \cap \mathcal{J}(F) \text{ is finite.} \quad (*)$$

If a Siegel disk or Herman ring existed, its boundary would be contained in the finite set in (*), since every Fatou-component boundary lies in the Julia set. This is impossible. To see it directly, an

open connected subset of the sphere with finite boundary is the sphere minus that boundary: after deleting the boundary, the subset is both open and closed, while the punctured sphere is connected. A sphere with finitely many punctures is not conformally a unit disk or a finite-modulus annulus (the one- and two-puncture cases are respectively $\mathbb{C}$ and $\mathbb{C}^*$, and three or more punctures have the wrong fundamental group).

Sullivan’s no-wandering-domain theorem now completes the proof. Every Fatou component eventually maps to a periodic one. The classification and the preceding exclusions leave only the immediate basins of the nineteen fixed roots. Hence every point of the original component converges to the root in that eventual immediate basin, and the root is common to the entire component. This proves Theorem 1.

Scope and reproducibility

The proof establishes only the $n = 18$ instance. It does not infer the answer for any other even n , does not classify the topology of the Julia set or of individual basins, and gives no claim about numerical convergence speed or stopping criteria. No broader priority claim is made beyond the comparison with the supplied literature above.

All map-specific sign decisions in Lemmas 2 and 3 use integers or rational numbers. The accompanying standard-library script `verify_orbit_threshold.py` recomputes these identities and inequalities without floating-point arithmetic. The mathematical conclusion rests on the displayed argument, not on a numerical experiment.

Language-model assistance was used to test algebraic steps, organize the proof, and prepare the manuscript. Automated exact-arithmetic checks and a separate proof-review pass were used as error-detection aids; they are not formal verification or independent peer review.

References

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